



Computer Networks

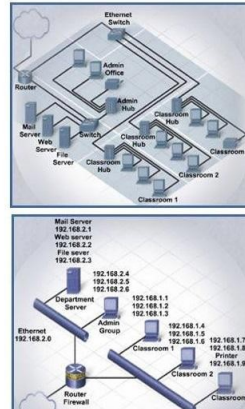
PPT-2: Network Topology

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Network Topology

Network topology refers to the physical or logical arrangement of nodes (devices) and the way they are interconnected within a computer network. It defines how data is transmitted between devices and the structure of the network.

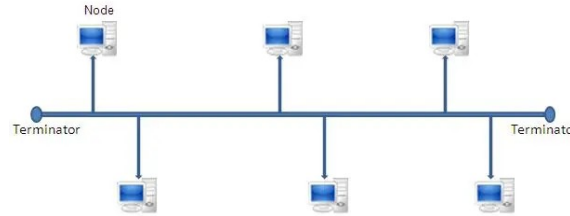
- Bus
- Ring
- Star
- Mesh
- Tree
- Hybrid



Physical topology is the physical layout of the components on the network.

Logical topology determines how the hosts access the medium to communicate across the network.

Bus



Advantages of Bus Topology

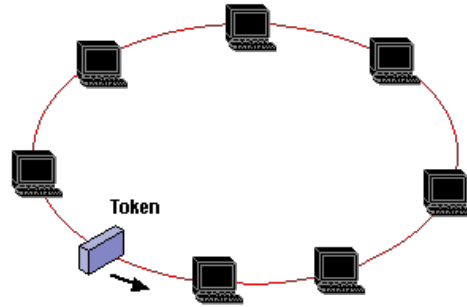
- Simplicity and Cost-Effectiveness
- Ease of Installation and Expansion
- Suitable for Small Networks
- Less Cable Usage

Disadvantages of Bus Topology

- Limited Scalability
- Single Point of Failure
- Performance Issues with Heavy Traffic
- Security and Privacy Concerns
- Limited Cable Length
- Not Suitable for High-Bandwidth Applications

Bus topology is most commonly associated with **IEEE 802.3 (Ethernet)**. Modern Ethernet still follows **IEEE 802.3**, but the **physical topology shifted to star** (using switches)

Ring



Advantages

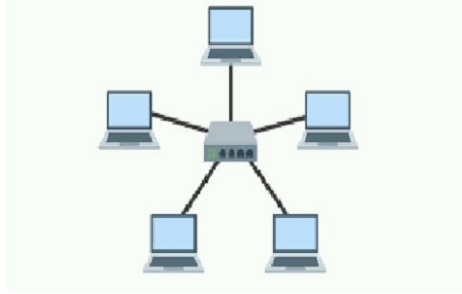
- Simplicity
- Equal Access to Resources
- Data Transmission Efficiency
 - (unidirectional - avoid collision)
- Ease of Expansion

Disadvantages

- Single Point of Failure
- Difficult Fault Identification
- Limited Scalability
 - Longer transmission time
- Slower Data Transfer Rates
 - If number of nodes increases
- Complex Network Management
- Less secure

IEEE 802.5 — Token Ring, **Developed by:** IBM, **Collision-free** (only the device with the token can transmit)

Star



Advantages

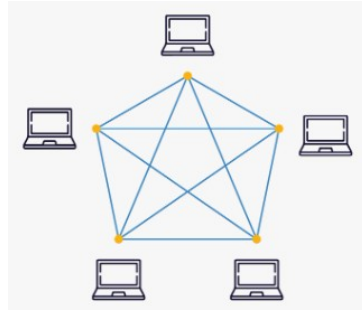
- Centralized Management
- Easy to Install and Expand
- Reliable Performance
- Isolation of Devices
- Scalability
- Better Performance in Heavy Traffic
- Security

Disadvantages

- Single Point of Failure
 - Dependency on Central Hub
- Cost of Installation
 - Addition central device
- Limited Scalability
 - Scalability Limits with Hub Capacity

IEEE 802.3 — (Modern Ethernet)

Mesh



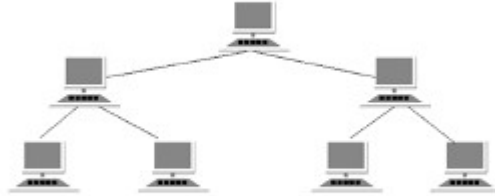
Advantages

- Redundancy and Reliability
- Fault Tolerance
- Supports Point-to-Point Communication
- Privacy and Security

Disadvantages

- Cost
- Complexity of Installation and Maintenance
- Limited Scalability
 - High Wiring Requirements

Tree



Advantages

- Easily scalable and expandable.
- Hierarchical structure for better organization.
- Fault isolation within branches.
- Efficient use of resources with a central hub.
- Centralized control for easier management.
- Easy to expand without disrupting the network.

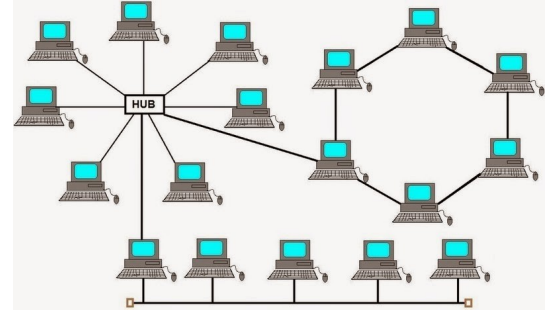
Disadvantages

- Dependent on the central root node.
- Can become complex to manage as it grows.
- Requires more cables, leading to higher costs.
- Performance may degrade with increased load.
- Expensive and time-consuming maintenance.
- Limited by the bandwidth of the backbone.

Hybrid

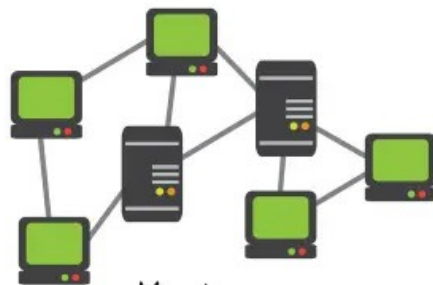
The purpose of implementing a hybrid topology is to leverage the strengths of multiple topologies while mitigating their individual weaknesses. By combining different topologies, a hybrid network can be customized to meet specific requirements and provide a more flexible and robust infrastructure.

- Improved Reliability
- Scalability
- Cost Efficiency
- Optimized Performance
- Enhanced Security

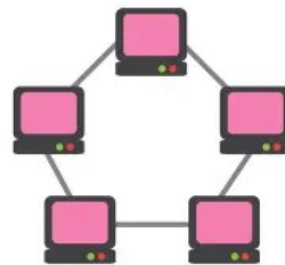




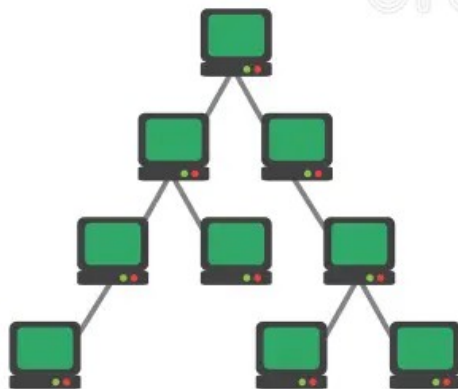
Star



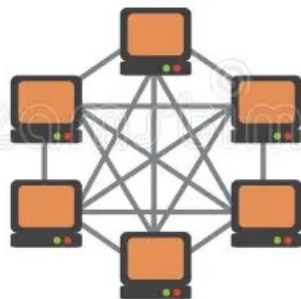
Mesh



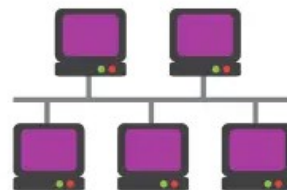
Ring



Tree



Fully Connected



Bus

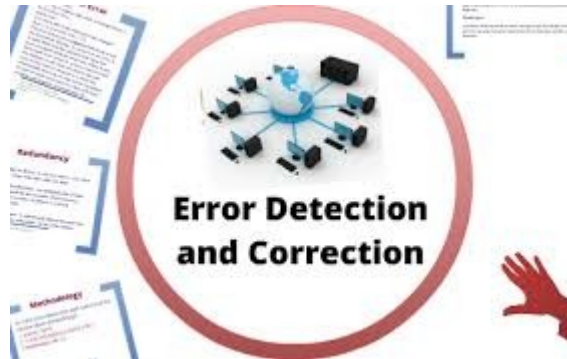


Line

Problems on Topology

Topology	Nodes (N)	No of Cables (NC)	No. of Ports per device (NOP)	Total No. of Ports (TNOP)
Bus	N	$N + 1$	1	N
Ring	N	N	2	$2 \times N$
Star	N	N	1	N
Mesh	N	$N(N-1)/2$	N-1	$N(N-1)$

Next Lecture





Computer Networks

PPT-2b: Network Topology 2b

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IEEE

IEEE (pronounced "Eye-triple-E")

IEEE stands for the Institute of Electrical and Electronics Engineers, the world's largest technical professional organization, dedicated to advancing technology and innovation.

IEEE 802.3 is the IEEE standard that defines **Ethernet**—the most widely used technology for wired local area networks (LANs).

IEEE 802.3 specifies:

- Physical layer (Layer 1)

- Data Link layer – MAC sublayer (Layer 2)



The 802.3 Frame Structure





IEEE